Erfan Shahabi

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Research Interests

Natural Language Processing, Large Language Models, Strategic Reasoning and Behavior of LLMs, Social Intelligence, Computational Social Science, Machine Behavior

Education

M.Sc in Information Technology Engineering, GPA: 4.00/4.00 (17.76/20) *Sep 2024 – Present*
University of Tehran, School of ECE *Tehran, Iran*

- Supervisors: Dr. Azadeh Shakery, Dr. Yadollah Yaghoobzadeh
- Thesis: An Investigation and Enhancement of Strategic Behavior of Large Language Models in Social Context

B.Sc in Industrial Engineering, GPA: 4.00/4.00 (18.07/20) *Sep 2019 – Aug 2023*
Islamic Azad University, Science and Research Branch *Tehran, Iran*

- Supervisor: Dr. Abbas Saghaei
- Thesis: Used Car Price Prediction Using Machine Learning Methods

Publications

- **SimInter: Evidence-Grounded User Simulation with Interview-Based Profile Enrichment for Conversational Recommender Systems**
  *Under Review at **NAACL 2027 Conference** (recommended for Findings in ARR May 2026 cycle)* | arXiv:2610.18308
  E Shahabi*, A Roshandel*, MS Sadeghi, A Shakery (*equal contribution, ordered randomly)
- **PerSo: Investigating Facets of LLMs' Social Intelligence in Real-World Persuasion Scenarios**
  *Under Review at **NAACL 2027 Conference*** | arXiv:2610.20548
  A Zamani*, S Esfahanian*, **E Shahabi**, MT Pilehvar, A Shakery, Y Yaghoobzadeh (*equal contribution, ordered randomly)

Professional Experience

Researcher *Sep 2024 – Present*
Intelligent Information Systems Lab *ECE @ University of Tehran, Tehran, Iran*

- Supervisors: Dr. Azadeh Shakery, Dr. Yadollah Yaghoobzadeh
- Research Area: Strategic Behavior and Social Reasoning of LLMs
- Developed **PerSo**, a benchmark evaluating social intelligence of LLMs using real-world persuasion scenarios from Reddit's Change My View (CMV) community
- Built a behavior-grounded user simulation framework (**SimInter**) for conversational recommender systems, leveraging audited synthetic interviews to enrich user profiles

AI Engineer / Data Scientist *Jul 2026 – Present*
Potk.ir *Tehran, Iran*

- Built data analysis and forecasting pipelines to predict market trends in electrical cable purchasing and sales
- Developed AI-powered tools leveraging LLMs and information retrieval techniques to support business decision-making

AI Engineer *Feb 2026 – Jul 2026*
Mofid Stock Brokerage *Tehran, Iran*

- Developed AI-powered financial tools leveraging LLMs and information retrieval techniques

Leadership Experience

Supervised 2 Bachelor Students for Research Project *Mar 2026 – Aug 2026*
Intelligent Information Systems Lab *ECE @ University of Tehran, Tehran, Iran*

- Supervised undergraduate research on LLM social intelligence evaluation
- Resulted in a paper submission to ACL ARR Oct 2026 cycle

Teaching Experience

Head Teaching Assistant, Intelligent Information Retrieval (Graduate) *Sep 2026 – Present*
Course Website · *Instructor: Dr. Azadeh Shakery* *ECE @ University of Tehran*

- Coordinated 6 computer assignments, workshops & the paper presentation module with a team of 6 TAs

Teaching Assistant, Natural Language Processing (Graduate) *Sep 2025 – Present*
Course Website · *Instructor: Dr. Heshaam Faili* *ECE @ University of Tehran*

- Designed TravelBot assignment: a conversational travel assistant using ReAct architecture and LangGraph for flight search, hotel booking, and trip planning
- Implemented RAG pipeline for retrieving destination information and generating personalized travel recommendations
- Led workshops mentoring students on LangChain and LangGraph frameworks for building agent-based systems

Teaching Assistant, Intelligent Information Retrieval (Graduate) *Sep 2025 – Sep 2026*
Course Website · *Instructor: Dr. Azadeh Shakery* *ECE @ University of Tehran*

- Designed assignments on conversational search, RAG systems, and multi-agent architectures
- Conducted workshops on probability theory and machine learning foundations for information retrieval
- Led technical workshops on LLMs in IR, covering LangChain, LangGraph, and ReAct patterns
- Guided students in implementing practical retrieval systems using state-of-the-art language models

Teaching Assistant, Advanced Data Mining (Graduate) *Feb 2026 – Present*
Instructor: Dr. Azadeh Shakery *ECE @ University of Tehran*

- Designed GNN-based assignment on academic network analysis, where students predict paper authorship and inter-paper relationships using graph neural networks
- Conducted hands-on workshops on GNN architectures and their applications in network data mining

Teaching Assistant, Natural Language Processing (Graduate) *Feb 2026 – Present*
Instructor: Dr. Yadollah Yaghoobzadeh *ECE @ University of Tehran*

- Designed and managed a financial advisory chatbot assignment, guiding students in building domain-specific conversational agents using LLMs
- Led workshops on fine-tuning and prompt engineering for task-specific NLP applications

Selected Projects

- Fine-Tuning LLaMA-3.2 using ORPO and DPO for Human-Preference-Based Alignment
- Quantization-Aware Reasoning and Text-to-SQL Agent Design using ReAct and LangGraph
- ChatFood: Building a ReAct-based Conversational Food chatbot with LangGraph and Gemini
- Conversational Search with RAG: Dataset Construction and Multifaceted Evaluation
- Deep Learning for Sequence Labeling and Text Summarization with RNNs and Attention
- Sentiment Classification of IMDB Reviews using Text Preprocessing and Classical IR Techniques
- Relevance Classification using BERT for Query-Document Pairs in IR Tasks
- Forecasting Community Dynamics in Social Graphs using Multi-Task GNNs

Technical Skills

- **Soft Skills:** Teamwork, Mentoring & Teaching, Meeting Tight Deadlines, Cross-cultural Communication
- **Programming Languages:** Python
- **Large Language Models:** Hugging Face, LangChain, LangGraph, LangFuse, Unsloth, PEFT, Datasets, TRL
- **Databases:** SQL, LanceDB
- **Machine Learning:** PyTorch, Scikit-Learn, Evaluate, TensorFlow
- **Retrieval:** Sentence Transformers, Lexical Retrieval, Semantic Retrieval, BM25

Honors and Awards

- Ranked 14 in Nationwide Graduate Level University Entrance Test of Information Technology Engineering

Languages

- **Persian:** Native **English:** IELTS Academic 7/9